

## **Kolla-Ansible: Complete Deployment guide (requirements → deploy → validate)**

Purpose: a ready-to-use, step-by-step document you can copy and run to deploy OpenStack with **Kolla-Ansible** (includes host/hardware/software/network requirements, example inventory and config files, common commands, post-deploy checks, and troubleshooting notes).

Sources: official Kolla-Ansible docs and project repos. ([OpenStack Docs](#))

### **Table of contents**

1. Quick overview
2. Requirements (hardware, OS, network, software)
3. Architecture options (all-in-one vs multinode)
4. Prepare control workstation (where you run Ansible)
5. Prepare target hosts (bootstrap tasks)
6. Example inventory (all-in-one and multinode)
7. /etc/kolla config files — globals.yml, passwords.yml (templates)
8. Install Kolla-Ansible (pip / from source)
9. Deploy commands (bootstrap → pull images → deploy → post-check)
10. Validation and smoke tests
11. Common adjustments and useful kolla-ansible variables
12. Backup, upgrade, and operational notes
13. Troubleshooting checklist & common errors
14. References

## 1) Quick overview

Kolla Ansible deploys OpenStack services in containers on your hosts using Ansible playbooks. It requires a control machine (where you run kolla ansible / ansible playbooks) and one or more target hosts (baremetal or VMs). The official quick start and user guides are the authoritative references. ([OpenStack Docs](#))

## 2) Requirements

### Host / hardware (minimum for evaluation / all-in-one)

- CPU: 2+ cores (more for production)
- Memory:  $\geq$  **8 GB** for an evaluation all-in-one (production needs much more).
- Disk:  $\geq$  **40 GB** free (evaluate); production requires larger volumes for images/volumes/DB).

These are evaluation minima, scale up for production. ([OpenStack Docs](#))

### Operating systems (supported)

- Commonly supported: **Ubuntu LTS, CentOS / RHEL** derivatives, and other distributions listed in the Kolla docs. Match the Kolla-Ansible release compatibility. ([OpenStack Docs](#))

### Network

- At least **2 network interfaces** recommended: one for management/internal traffic and one for external (public) traffic / provider network. Use a consistent IP plan; configure DNS or /etc/hosts entries for all hosts. Floating IPs and an external network are required for public access to instances. ([OpenStack Docs](#))

### Software

- **Python 3.x** on control and target hosts.
- **Ansible**: compatible major version per Kolla release (check release notes). Typically, Ansible 4+ for modern Kolla releases. ([OpenStack Docs](#))
- **Container runtime** (Docker or other supported container runtime) on target hosts; Kolla deploys services as containers. ([OpenStack Docs](#))
- pip3, git, python3-venv recommended on the control host.

## Access

- SSH connectivity from control host to all target hosts. Passwordless SSH keys are **highly recommended** for Ansible. Ansible can use passwords but keys are preferred. ([OpenDev: Free Software Needs Free Tools](#))

## 3) Architecture options

- **All-in-one:** single host runs all OpenStack containers, for evaluation only.
- **Multinode:** (separate roles) control (API/db/message), compute (nova compute), storage (cinder, swift), network (neutron) nodes. Use inventory to map hosts to roles. See production architecture guide for clustering and HA patterns. ([GitHub](#))

## 4) Prepare control workstation (where you run Ansible / kolla-ansible)

Run these commands on your control machine (example uses Rocky Linux):

```
# --- Install system dependencies ---
sudo dnf -y update
sudo dnf -y install epel-release
sudo dnf -y install python3 python3-pip python3-virtualenv
python3-devel git gcc openssl-devel libffi-devel

# --- Create a virtual environment (recommended) ---
python3 -m venv ~/kolla-venv
source ~/kolla-venv/bin/activate

# --- Clone the Kolla-Ansible repository ---
git clone https://github.com/openstack/kolla-ansible.git
cd kolla-ansible

# (Optional) Checkout a stable release branch - example for
2024.1:
# git checkout stable/2024.1

# --- Install Python requirements and Kolla-Ansible itself --
-
pip install -U pip wheel
pip install -r requirements.txt
pip install .
```

## Notes:

- You can also pip install kolla-ansible from PyPI for released versions, but cloning lets you choose branches. ([GitHub](#))

## 5) Prepare target hosts (bootstrap tasks)

On each target host (control/compute/storage):

```
# --- Basic packages (Rocky Linux version) ---
sudo dnf -y update
sudo dnf -y install epel-release
sudo dnf -y install openssh-server python3 python3-pip sudo
rsync ca-certificates curl

# Enable and start SSH service (if not already running)
sudo systemctl enable --now sshd

# --- Create a user for Ansible / Kolla operations ---
sudo useradd -m -s /bin/bash kolla

# Set a password (optional, if not using SSH keys)
# sudo passwd kolla

# Copy your SSH public key for passwordless access
# (Run this from the control node after generating an SSH
key)
# ssh-copy-id kolla@<target-host>

# --- Grant passwordless sudo privileges to the 'kolla' user
---
echo "kolla ALL=(ALL) NOPASSWD:ALL" | sudo tee
/etc/sudoers.d/kolla
sudo chmod 440 /etc/sudoers.d/kolla

# --- Install and enable container runtime (Docker example) -
--
sudo dnf -y install docker
sudo systemctl enable --now docker

# Add the 'kolla' user to the docker group so it can manage
containers
sudo usermod -aG docker kolla

# Verify Docker is working
sudo docker run hello-world || echo "Docker test container
executed"
```

Other items:

- Ensure hostnames and /etc/hosts or DNS are set so each host resolves properly.
- Time sync (ntp / chrony) recommended.

- Allow traffic on necessary ports (Ansible uses SSH, OpenStack services will expose ports later). ([OpenStack Docs](#))

## 6) Example Ansible inventory

### Multinode (simple 3-node example)

```
cat ~inventory/multinode:
[control]
controller1 ansible_host=192.0.2.11
controller2 ansible_host=192.0.2.12
controller3 ansible_host=192.0.2.13

[network]
controller1
controller2
controller3

[compute]
compute1 ansible_host=192.0.2.21
compute2 ansible_host=192.0.2.22

[storage]
storage1 ansible_host=192.0.2.31
```

Modify groups and hostnames to reflect your deployment. Kolla ships sample inventory files, adjust them for your environment. ([OpenDev: Free Software Needs Free Tools](#))

## 7) /etc/kolla configuration files (copy-paste templates)

Create /etc/kolla and put these files there.

```
sudo mkdir -p /etc/kolla
sudo chown $USER:$USER /etc/kolla
```

### A) globals.yml (minimal example — copy & edit)

Many more variables are available, below are the most commonly changed items. See the etc/kolla/globals.yml reference for full options. ([OpenDev: Free Software Needs Free Tools](#))

```
# --- Kolla-Ansible Global Config (Rocky Linux Example) ---

kolla_base_distro: "rocky"
kolla_install_type: "binary"
openstack_release: "2024.1"
```

```
kolla_internal_vip_address: "10.0.0.100"
kolla_external_vip_address: "192.168.0.100"
network_interface: "eth0"
neutron_external_interface: "eth1"
enable_haproxy: "yes"
enable_cinder: "yes"
enable_neutron_provider_networks: "yes"
enable_heat: "no"
kolla_ansible_user: "{{ lookup('env', 'USER') }}"
```

## B) passwords.yml

You **must** provide secure passwords. For testing you can generate them with kolla-genpwd or scripts in the repo. Below is an example skeleton, **do not** use weak passwords in production.

```
keystone_admin_password: "REPLACE_WITH_STRONG_PASSWORD"
keystone_admin_token: "REPLACE_IF_USED"
admin_password: "REPLACE_WITH_STRONG_PASSWORD"
mysql_root_password: "REPLACE_WITH_STRONG_PASSWORD"
rabbitmq_password: "REPLACE_WITH_STRONG_PASSWORD"

# many more keys, use kolla tools to generate the full file
# or copy from examples
```

Generate recommended password file:

```
# kolla-ansible provides a helper (if installed)

kolla-genpwd

# this will populate /etc/kolla/passwords.yml with generated
# passwords
```

(If kolla-genpwd is not available in your install method, see the repo scripts or generate passwords yourself.) ([OpenStack Docs](#))

## 8) Install Kolla-Ansible (detailed)

Two common approaches:

### A) Install from PyPI (released package)

```
# --- Install Kolla-Ansible via PyPI (inside your venv) ---
pip install -U pip wheel
pip install kolla-ansible

# --- Create configuration directories ---
sudo mkdir -p /etc/kolla
```

```

sudo chown $USER:$USER /etc/kolla

# --- Copy example configuration and inventory templates ---
# (these come with the installed Python package)
cp -r $(python3 -c "import kolla_ansible;
print(kolla_ansible.__path__[0])")/etc/kolla/* /etc/kolla/

# Create an inventory directory for your setup
mkdir -p ~/inventory

# (Optional) Copy provided example inventories
cp /usr/local/share/kolla-ansible/ansible/inventory/*
~/inventory/ 2>/dev/null || true

```

### Notes for Rocky Linux:

- Files from kolla-ansible are installed under /usr/local/share/kolla-ansible/ansible/.
- You can verify installation with:

### B) Install from source (recommended for choosing branches)

```

# --- Clone the official Kolla-Ansible repository ---
git clone https://github.com/openstack/kolla-ansible.git
cd kolla-ansible

# (Optional) Checkout a stable release branch
# Example: git checkout stable/2024.1

# --- Install requirements and Kolla-Ansible itself ---
pip install -U pip wheel
pip install -r requirements.txt
pip install .

# --- Copy configuration templates to /etc/kolla ---
sudo mkdir -p /etc/kolla
sudo chown $USER:$USER /etc/kolla
cp -r etc/kolla/* /etc/kolla/

# --- Prepare your inventory folder ---
mkdir -p ~/inventory
cp -r ansible/inventory/* ~/inventory/ 2>/dev/null || true

```

### Verification

After either installation method, confirm it works:

```
# Should display version info
kolla-ansible --version

# List available Ansible playbooks
ls /usr/local/share/kolla-ansible/ansible/ 2>/dev/null || ls
ansible/
```

## 9) Deploy commands (bootstrap → pull → deploy)

Run from your control host (in the virtualenv where kolla-ansible is installed). Use the inventory you prepared.

### 1. Bootstrap hosts (installs docker, creates users, configures kernel, etc.)

```
# example, using inventory file at 'inventory/multinode'
kolla-ansible -i inventory/multinode bootstrap-servers
```

### 2. Prechecks / hosts prep

```
kolla-ansible -i inventory/multinode prechecks
kolla-ansible -i inventory/multinode pull
```

pull will fetch container images required for the configured OpenStack release.

### 3. Run deploy playbooks

```
kolla-ansible -i inventory/multinode deploy
```

### 4. Post-deploy (configure admin openrc)

```
# Generate admin-openrc.sh
kolla-ansible post-deploy
# Then source the file created in /etc/kolla or check docs
for exact path
source /etc/kolla/admin-openrc.sh
```

Important: for development quickstart there are simplified scripts; for production you may want to use the full recommended sequence in the quickstart guide. ([OpenStack Docs](#))

## 10) Validation and smoke tests

After post-deploy:

```
# source admin credentials (path from post-deploy)
source /etc/kolla/admin-openrc.sh

# check keystone services / token
openstack token issue
```

```
# list services
openstack service list

# try creating small temp resources (flavor, network, boot a
tiny instance)
openstack flavor create --id 0 --ram 64 --disk 1 --vcpus 1
tiny
```

You can also use kolla-ansible provided healthchecks and monitoring playbooks; see operating docs. ([OpenStack Docs](#))

## 11) Common variables & useful tweaks

(Examples you may change in /etc/kolla/globals.yml)

- openstack\_release - must match the images you pull / desired release. ([OpenDev: Free Software Needs Free Tools](#))
- kolla\_install\_type: source|binary - choose whether to run services from source or prebuilt packages.
- docker\_namespace and kolla\_internal\_vip\_address - set to control image names and cluster VIPs.
- neutron\_plugin\_agent: openvswitch - common default; set to linuxbridge if you prefer.  
Refer to etc/kolla/globals.yml for the complete list. ([OpenDev: Free Software Needs Free Tools](#))

## 12) Backup, upgrade, and operational notes

- Keep backups of /etc/kolla/globals.yml, /etc/kolla/passwords.yml, and your Ansible inventory - these are critical.
- For upgrades: follow Kolla-Ansible release notes and upgrade procedures; test in a staging environment first. ([OpenStack Docs](#))

## 13) Troubleshooting checklist & common errors

- **Ansible SSH failures:** ensure SSH keys, ansible\_user/kolla\_ansible\_user match, and sudo permissions exist. (Ansible will log the failing task.) ([OpenDev: Free Software Needs Free Tools](#))

- **Image pull errors:** ensure the control host and target hosts can reach the container registry; consider mirroring images to a private registry if bandwidth constrained.
- **Service start failures (containers exiting):** docker logs <container> on host; check /var/log/kolla and journalctl for system issues.
- **Network issues:** validate host routing, bridge interfaces, and MTU; check neutron logs inside containers.
- **Mismatched versions:** ensure Ansible, Docker, and OS versions are compatible with chosen Kolla-Ansible release. Consult release notes. ([OpenStack Docs](#))

#### 14) References (read first)

- **Kolla-Ansible - Quickstart & User guides** (main docs, step-by-step): ([OpenStack Docs](#))
- **Kolla-Ansible project (GitHub)** (source, README): ([GitHub](#))
- **Ansible deployment notes and inventory guidance** (repo docs): ([OpenDev: Free Software Needs Free Tools](#))
- **Globals.yml reference (variables)** (opendev): ([OpenDev: Free Software Needs Free Tools](#))
- **Operating Kolla (post-deploy, upgrades, maintenance):** ([OpenStack Docs](#))